

THE SCOPE OF THE TOPIC

The proposed system is an internal technical interview preparation platform designed specifically for university students in Computer Science, Software Engineering, and related fields. It aims to prepare undergraduates, placement year seekers, and fresh graduates for rigorous technical interviews.

- **Similar Products:** LeetCode, HackerRank, NeetCode, AlgoMonster.
- **System Platform:** Hosted on cloud infrastructure (e.g., AWS/GCP) to ensure remote accessibility for students practicing from home or on campus.
- **Advanced Features:** Integration with university Single Sign-On (SSO) and student email verification to strictly limit access to internal students. An integrated code execution engine capable of running student submissions against predefined test cases.
- **Data Requirements:** A relational database is required to manage student profiles, track individual progress, store categorical coding challenges (e.g., Hash Maps, Sliding Window), and host community-driven solutions and editorials.
- **User Base:** Primarily targeted at students seeking internships, with faculty and staff managing content and monitoring progress.

Task 2. Stakeholder Analysis

The following outlines the stakeholders based on their level of interaction with the system, mapping direct users, indirect users, and other vested parties for the Onion Diagram.

Ring 3: Other Stakeholders (Outer Layer)

- External Tech Recruiters
- University Management & Board
- Cloud Infrastructure Providers

Ring 2: Indirect Users (Middle Layer)

- University Career Services Team
- CS Lecturers & Module Leaders

Ring 1: Direct Users (Inner Layer)

- CS & SE Students

- Platform Administrators
- Content Managers (TAs/Faculty)

System Core:

- University Interview Prep Platform

Task 3. Requirements Elicitation

Identified Elicitation Methods:

- **Direct Users (Students):** One-on-one structured interviews to understand pain points with existing public platforms and identify specific preparation needs.
- **Indirect Users (Career Services/Lecturers):** Targeted questionnaires to gather requirements on analytics, reporting dashboards, and curriculum alignment.

Designed Interview Questions:

- **For Students:** "What features do you feel are missing from platforms like LeetCode when preparing for placement years?" / "How important is seeing university-specific editorial solutions to you?"
- **For Faculty:** "What metrics would be most useful to track regarding student performance in algorithmic problem-solving?"

AI Simulation

Simulation 1: The Direct User (Student)

- **Prompt to AI:** "Act as a third-year Computer Science student preparing for a placement year. What are your core requirements for a university-hosted NeetCode-style interview prep platform?" PDF + 1
- **AI Response:** "As a third-year CS student, my primary goal is to pass technical coding interviews. My core requirements for the platform are:
 1. **Curated Problem Sets:** The platform must categorize problems by topic (e.g., arrays, binary search) and difficulty so I can follow a structured study path.
 2. **Integrated Code Execution:** I need to be able to write, compile, and execute my code directly in the browser against hidden test cases to verify my solutions.
 3. **High-Quality Editorials:** When I get stuck, I need clear, step-by-step editorial explanations, ideally with time and space complexity breakdowns.
 4. **Community Solutions:** I want to see how other students solved the same problem to learn different approaches and language-specific tricks.
 5. **Progress Tracking:** A dashboard showing how many problems I've solved and what areas I need to practice more."

Simulation 2: The Direct User (System Administrator)

- **Prompt to AI:** "Act as a University IT Administrator. What are your core system requirements for securely hosting this new internal coding platform?" PDF + 1
- **AI Response:** "As a System Administrator, my focus is on security, stability, and compliance. My core requirements are:
 1. **Access Control:** The system must strictly enforce university email verification, ideally integrating with our existing Single Sign-On (SSO) infrastructure.
 2. **Database Security:** The relational database storing user profiles and submissions must be encrypted at rest and in transit.
 3. **Cloud Scalability:** The cloud architecture must be able to auto-scale during peak usage times (e.g., right before placement application deadlines).
 4. **Admin Dashboard:** A centralized portal to manage user access, monitor server health, and manage the underlying infrastructure."

Task 4. Requirements Analysis

Based on the elicitation phase, the requirements are analyzed across two distinct levels:

User-Level Requirements

- Students require intuitive categorization of coding problems and immediate feedback from the code execution engine.
- Faculty and Career Services require read-only dashboards to monitor aggregate student readiness.

System-Level Requirements

- The system must securely authenticate users via the university email domain.
- The backend must safely sandbox and execute arbitrary code submissions in multiple languages (Python, Java, C++) within a 2-second timeout window.
- The relational database must maintain referential integrity between users, submissions, and problems.